Marc Schalberger

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WORK EXPERIENCE

Statistical Consultant 2023 - today Statistical Consulting Unit fu:stat, Freie Universität Berlin

Data Analyst 2020 - 2021 Vehiculum Berlin

Research Assistant 2023 Geschwister Scholl Institute, Ludwig Maximilian University of Munich

Student Assistant, Math Tutor 2018 - 2019 European University Viadrina

EDUCATION

Ph.D. in Statistics 2023 - 2026 Freie Universität Berlin

M.Sc. in Statistics & Data Science 2021 - 2023 Ludwig Maximilian University of Munich

Visiting Student 2020 Sophia University Tokyo

B.Sc. in Business Administration 2017 - 2020 European University Viadrina

LANGUAGE SKILLS

German Native
English Fluent

TEACHING EXPERIENCE

Courses

- Causal Machine Learning (2025 and 2026, Teaching Assistant, graduate level)

Workshops

- Foundations of Statistics (1-day)
- Introduction to Python (2-day)
- Introduction to R (2-day)
- Introduction to Statistical Learning (1-day)

Tutorials

- Statistical Modeling of Signed Networks with ergm.sign, 3-hour workshop on the R package ergm.sign in Linköpping August 2026 (with Cornelius Fritz)

AWARDS AND GRANTS

- Early Career Travel award, American Statistical Association (ASA), 2025
- Travel Grant, Deutsche Statistische Gesellschaft (DStatG), 2025
- PROMOS Scholarship, Deutscher Akademischer Austauschdienst (DAAD), 2020

RESEARCH

Publications under Review

[5] Schalberger, M., & Fritz, C. (2025+). Scalable Signed Exponential Random Graph Models under Local Dependence. https://doi.org/10.48550/arXiv.2507.07660

[4] Razpotnik, M., ..., Schalberger, M., … (2026) Machine Learning-Based Decision Support for Step-Up Therapy After Endosonography-Guided Drainage of Pancreatic Fluid Collections: Multicenter Development and Validation Study

Working papers

[3] Schalberger, M., Mehrl, M., Fritz C. (2026) Order in Motion: Structural Balance in International Relations since 1900

[2] Schalberger, M., Fritz C., Krivitsky, P., Mehrl M. (2026) ergm.sign: Modeling Signed Networks in R

[1] Schalberger, M. (2026) A Survey of Statistical Models for Dynamic Signed Networks

Theses

- *Exponential Random Graph Models for Signed Networks: Implementation and Application*. Master Thesis, LMU Munich (2023)
- *Forecasting Stock Prices using ARIMA and GARCH model.* Bachelor Thesis, European University Viadrina (2020)

Software

- ergm.sign: R package on CRAN to estimate, simulate, and assess the fit of Signed Exponential Random Graph Models.
- bigsergm: R package on GitHub providing a toolbox to analyze and simulate large signed networks under local dependence.

Conference Talks (mm-yyyy)

- **08-2026 (Linköping, SE):** Order in Motion: Structural Balance in International Relations since 1900. *European Conference on Social Networks 2026*
- **09-2025 (Edinburgh, UK)**: Scalable Estimation of Signed Exponential Random Graph Models with Local Dependence. *RSS International Conference 2025*
- **08-2025 (Nashville, USA)**: Scalable Estimation of Signed Exponential Random Graph Models with Local Dependence. *Joint Statistical Meeting 2025*
- **06-2024 (Edinburgh, UK)** Scalable Estimation of Signed Exponential Random Graph Models with Local Dependence. *Sunbelt 2024*

Reviewer of research papers (in alphabetical order)

Econometric Reviews, Journal of the Royal Statistical Society Series A

Professional memberships

Member of the German Statistical Society (Deutsche Statistische Gesellschaft), American Statistical Association and Royal Statistical Society.